Cats vs. Dogs II

Convolutional Classifiers Beyond Internal Validation

Mateus Bandeira • Theo Veiga

al.mateus.torres@impatech.edu.br • al.theo.ambrosio@impatech.edu.br

Advisor: Lucas Nissenbaum

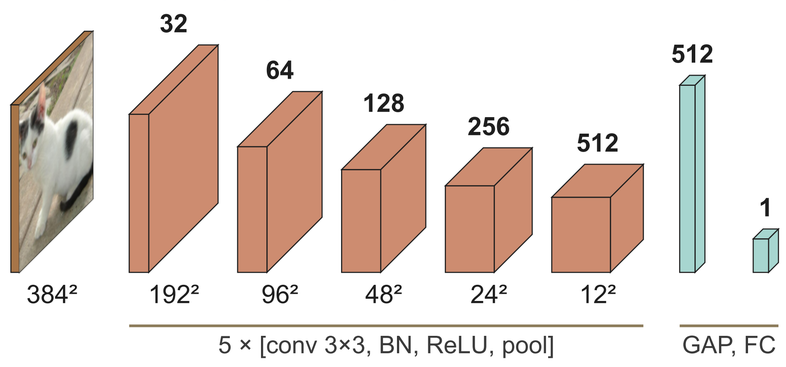
Instituto de Matemática Pura e Aplicada

Introduction

In Part I we explained how convolutional neural networks learn to classify images. In this poster we look at what happens in practice. Four students built their own networks for the IMPA Tech Cats vs. Dogs competition, all trained on the 25,000 labeled photos of Kaggle's Dogs vs. Cats [3], but each with its own architecture, preprocessing and training method.

Each student split the Kaggle photos into training and validation sets in a different way, so their validation scores are not directly comparable. To compare the models on equal terms, all four were tested on the same new set of **78 photos of pets from the IMPA Tech community (45 cats and 33 dogs)**, which no model saw during training. We describe the architectures and training methods, compare the results, and discuss which choices may explain the differences.

Architecture example



Training curves

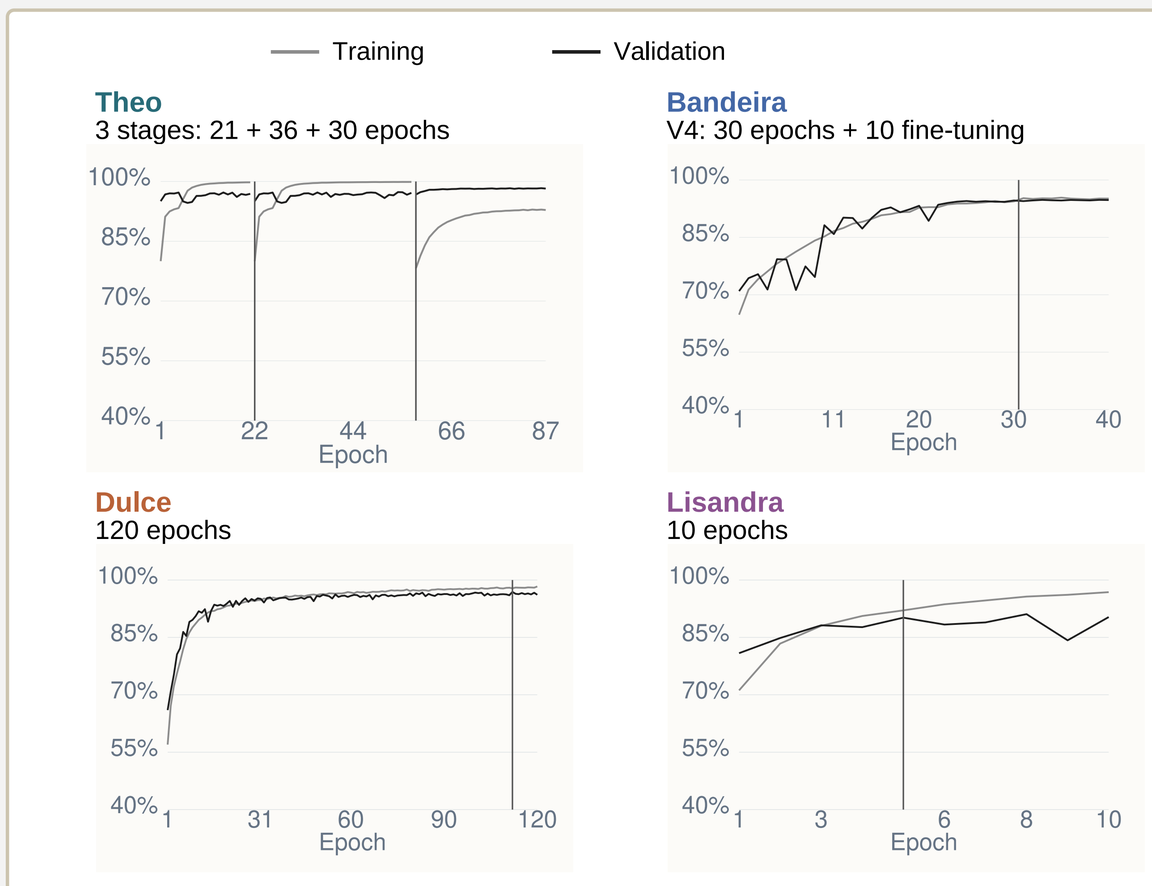


Figure 1 – Training and validation accuracy; vertical lines mark stages or the chosen epoch.

Training methods

Theo: Aug8, 8 views per image



Bandeira: light, online **Dulce: fixed + online** **Lisandra**



Figure 2 – One Kaggle training photo under each student's data augmentation.

	Theo	Bandeira	Dulce	Lisandra
Initialization	ImageNet	from scratch	from scratch	from scratch
Fine-tuning	whole net (Aug8 stage)	blocks 4–5 + head	no	no
Optimizer	Adam, then AdamW + cosine	AdamW + cosine	Adam	Adam
Dropout	10% before FC	50% before FC	20% per conv block	50% per hidden FC
Epochs	21 + 36 + 30	30 + 10	120	10 (kept 5)
Batch	64	64	32	50
Augmentation	Aug8 (8 views)	light, online	fixed + online	none
Checkpoint	best val. acc.	lowest val. loss	best val. acc.	lowest val. loss

Each model was tested with the preprocessing it was trained with: Theo, Bandeira and Dulce crop the center of the photo and keep its proportions, while Lisandra squeezes it into a square. Squeezing Bandeira's inputs instead would drop it from 64 to 61 correct photos.

The networks also end differently. Theo and Bandeira average each channel (global average pooling) before a single output unit, leaving only 257 and 513 parameters in the head. Dulce and Lisandra flatten the maps into large fully connected layers, which hold 79% and 94% of their parameters.

Examples from the Pets photos



Figure 3 – Top: a correct prediction; bottom: an error (true → predicted class).

Results on the community pets

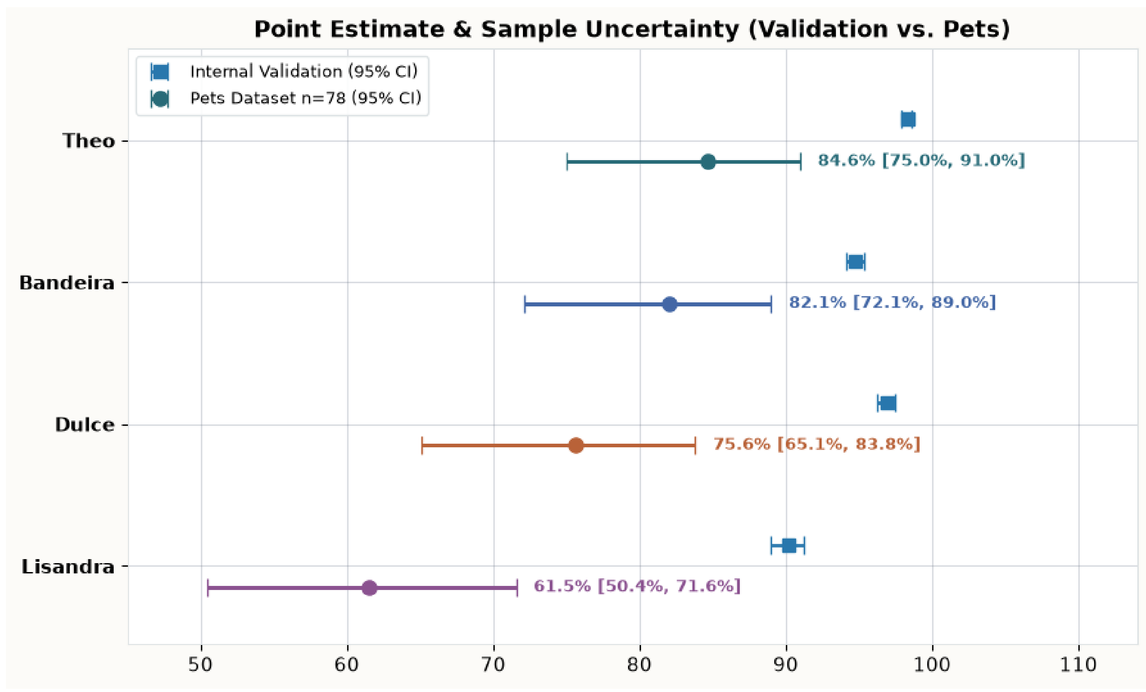


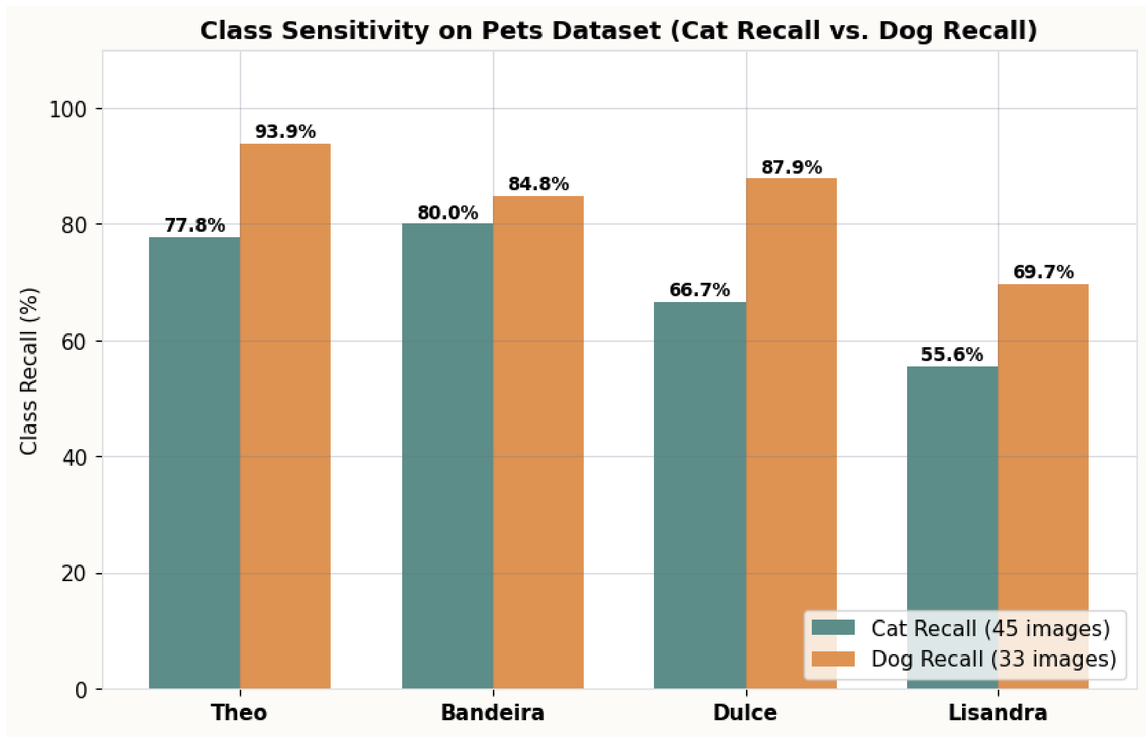
Figure 4 – Accuracy with Wilson 95% intervals: 78 photos leave wide intervals.

Performance summary

Rank	Model	Input regime	Correct /78	Acc.	Bal. acc.	Cat recall	Dog recall	Drop vs. val.
1st	**Theo** (ResTheo34 Aug8)	Direct resize 224 (ImageNet pretraining)	66/78	84.62%	85.86%	35/45 (77.8%)	31/33 (93.9%)	−13.70 p.p.
2nd	**Bandeira** (CNN V4 Final)	Proportional crop 384 (native)	64/78	82.05%	82.42%	36/45 (80.0%)	28/33 (84.8%)	−12.71 p.p.
3rd	**Dulce** (CNN224 v2)	Proportional box crop 224 (native)	59/78	75.64%	77.27%	30/45 (66.7%)	29/33 (87.9%)	−21.26 p.p.
4th	**Lisandra** (AlexNetBinary)	Direct resize 227 (native)	48/78	61.54%	62.63%	25/45 (55.6%)	23/33 (69.7%)	−28.62 p.p.
Mean	Mean of 4 models	—	59.25/78	75.96%	77.05%	31.5/45 (70.0%)	27.75/33 (84.1%)	−19.07 p.p.

Tabela 1 – All models on the 78 community photos, each with its own preprocessing.

Class sensitivity on the community pets



Conclusions

All models lost accuracy on the community photos; most errors are cats predicted as dogs. The models differ in many choices at once, so the reasons below are hypotheses.

Theo (66/78) is the only model pretrained on ImageNet (over a million varied photos), and Aug8 adds noise and occlusions.

Bandeira (64/78, interval overlapping Theo's) came close from scratch, with the smallest network (1.6M parameters), the largest input and a fine-tuning stage.

Dulce and **Lisandra** had the largest drops (21 and 29 points); both end in large fully connected heads, and Lisandra trained the shortest, without augmentation. At last, we ask you to see our repository with all the results on the QR code below.

References and reproducibility

[1] HE, K.; ZHANG, X.; REN, S.; SUN, J. *Deep Residual Learning for Image Recognition*. CVPR, 2016, p. 770–778.

[2] PRINCE, S. J. D. *Understanding Deep Learning*. Cambridge, MA: MIT Press, 2023.

[3] KAGGLE. *Dogs vs. Cats*. Competition dataset, 2013. Available at: https://www.kaggle.com/c/dogs-vs-cats.

[4] KRIZHEVSKY, A.; SUTSKEVER, I.; HINTON, G. E. *ImageNet Classification with Deep Convolutional Neural Networks*. NeurIPS, 2012, p. 1097–1105.

[5] WILSON, E. B. Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, v. 22, n. 158, p. 209–212, 1927.

[6] PICME Cats vs. Dogs. *Unified repository, published evaluation and analysis notebook*. Commit 67e51de, 2026.

Models: Dulce Costa, Lisandra Fagundes, Mateus Bandeira, Theo Veiga. **AI Disclosure:** ChatGPT, Claude and Gemini were used to assist with layout, figures, editing and grammar.